Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/10/2024
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jimq@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3993
Organization Name:	University of Maryland		
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PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2015	**End Date:**	02/08/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 02/08/2025 per NSSC information (Ed., 3/10/24) NOTE: Extended to 02/08/2024 per NSSC information (Ed., 2/24/23) NOTE: Extended to 02/08/2023 per NSSC information (Ed., 4/11/22) NOTE: Extended to 02/08/2022 per NSSC information (Ed., 12/3/21) NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 11/7/19) NOTE: Extended to 12/31/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	Grant 80NSSC20M0119 has been extended to 02/08/2026 under new PI, Peter Sunderland (Ed., 4/15/25).		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) deRis, John Ph.D. (FM Global)		
Grant/Contract No.:	80NSSC20M0119 ; NNX15AD06A		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	2025 UPDATE: Grant 80NSSC20M0119 has been extended to 02/08/2026 under a new Principal Investigator (PI): Peter Sunderland. This is the final record under Dr. Quintiere. For current reporting, see the new project record at: https:// . (Ed., 4/15/25). NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," grant NNX10AD98G, with the same Principal Investigator (PI) (Ed., 1/14/2015) A novel experimental apparatus, the Burning Rate Emulator (BRE), was developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of virtual solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it enables the emulation of steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements were conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station. The Burning Rate Emulator (BRE) simulated the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE controlled the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This allowed it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment was considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) were considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows. 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	We continued working on the data analysis and publication of all the Burning Rate Emulator (BRE) data. One journal paper and one conference abstract were published this year. [Ed. Note: See Cumulative Bibliography and previous Fiscal Year reports for additional information (1/16/24).]
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Articles in Peer-reviewed Journals	Dehghani P, Quintiere JG. "Flammability maps for microgravity burning of a small flat material in a quiescent ambient." Combust Flame. 2023 Jul 1;253:112809. https://doi.org/10.1016/j.combustflame.2023.112809 , Jul-2023
Papers from Meeting Proceedings	Dehghani P, Sunderland PB, Quintiere JG, deRis J, Baum HR. "Using a gas supplied burner to obtain flammability maps of solids and liquids in microgravity in a quiescent ambient without external radiation." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023